Heterogeneous Treatment Effect Estimation with Instrumental Variable Methods

Journal Title
XX(X):1–11
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DOI: 10.1177/ToBeAssigned
www.sagepub.com/
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Amir Aamodt Kazemi[1,2,3], Joseph Sexton[2] and Inge Christoffer Olsen[1,3]

Abstract

Instrumental variable (IV) methods are widely used to address unmeasured confounding in observational studies but typically estimate the local average treatment effect (LATE) for compliers. If there exist effect modifying variables that also affect the choice of treatment, the LATE may have limited clinical relevance. To address this limitation, we propose a novel framework for estimating the conditional average treatment effect (CATE) for a categorical treatment variable in the presence of both observed confounding and effect-modifying variables in addition to valid IVs. Building on recent developments in data fusion between randomized trial and observational data, we combine IV estimators with estimators based on covariate adjustment. By making parametric assumptions about the residual unobserved confounding effect, we obtain a consistent estimator of the CATE with relatively high efficiency. In a simulation study with nonlinear effect modification and unobserved confounding, we show that our estimator outperforms competing IV and covariate-adjusted estimators in terms of mean squared error. We further illustrate the applicability of the method, utilizing it to compare tumor necrosis factor inhibitors as the first line biologic treatment of rheumatoid arthritis conditional on age. The analysis identifies modest age-related heterogeneity in treatment effects and provides population-relevant clinically interpretable estimates. This work provides a general framework for combining instrumental and confounding information to estimate individualized causal effects when treatment effects are heterogeneous and there are more than two treatment alternatives.

Keywords

Conditional Average Treatment Effect, Combining Information, Causal Inference

1 Introduction

Instrumental variable (IV) methods have gained much popularity in medical research in recent years due to their ability to circumvent the assumption of no unobserved confounding effects in observational causal inference. The average treatment effect (ATE) can be estimated with IV methods only if one can assume homogeneity, meaning that there exist no effect modifying variables that also affect the choice of treatment[1]. For heterogeneous treatment effects, IV methods estimate the local average treatment effect (LATE) for compliers, where compliance is defined using counterfactual treatment choices[2]. Therefore, one can argue that compliance is not an observable variable, even in principle. This results in IV methods estimating the treatment effect for an unidentifiable subpopulation[3].

In randomized controlled trials (RCTs), which are the gold standard for measuring causal effects, the intervention applied in the study and the intervention proposed after the study are the same. In contrast, this does not need to be the case in an IV analysis, namely, the intervention proposed after the study can be on another variable than the IV[4]. If there exist effect modifiers that can affect the choice of treatment, the relevance of standard IV estimates to guide clinical decision making depends on how closely the IV mimics the intervention of interest[5]. The LATE might be an interesting target parameter if the intervention suggested affects decision makers in a similar way to the IV, in which case the LATE can be interpreted as the per-protocol effect of an RCT that applies that intervention[6]. However, if the intervention of interest is not similar to the IV, the LATE might be an uninteresting target parameter. For example, the price of commodities such as cigarettes, sugar and alcohol can be used as an IV to estimate their causal effect on health outcomes. Based on the results of such an analysis, an intervention can be suggested introducing additional taxes on the commodity of interest, likely increasing the price. In this case, one could argue that the subpopulation whose choice of exposure is affected by this intervention is the same as the complier subpopulation in the IV analysis. As a contrast example, the effect of a health service on a health outcome can be estimated using distance to the nearest hospital providing that health service as an IV. However, if the intervention of interest is to introduce subsidies for that health service, one can argue that the effect for the complier subpopulation in the IV analysis, whose choice of treatment is affected by distance, could be different from the effect for

[1]Department of Research Support for Clinical Trials, Oslo University Hospital, Klaus Torgårds vei 3, 0372, Oslo, Norway
[2]Department of Rheumatology, Diakonhjemmet Hospital, Diakonveien 12, 0370, Oslo, Norway
[3] Oslo Centre for Biostatistics and Epidemiology, Faculty of Medicine, University of Oslo, Sognsvannsveien 9, 0372, Oslo, Norway

Corresponding author:
Amir Aamodt Kazemi
Email: amirhosk@uio.no

the subpopulation whose choice of treatment is affected by price.

Homogeneity enables generalization of effect estimates from one subpopulation to another. However, it might be a strong assumption in many cases. When the treatment effect is heterogeneous, one should observe the effect modifying variables that affect the choice of treatment and adjust for them. Then, an interesting target parameter is the conditional average treatment effect (CATE). In CATE estimation, the treatment effect is estimated as a function of observed effect modifiers. The estimated function can then be applied for individualized clinical decision-making. However, in practice, IV analysis alone might struggle to estimate the CATE in finite samples, because there might be regions of effect modifiers that exhibit little to no compliance to the instrument, leading to unstable estimates. A solution is to combine covariate and instrumental information.

Several methodological papers have been published providing a framework for combining information from RCTs and observational data to estimate the CATE[7]. The main idea is to utilize RCTs with support in a limited region of the effect modifier space to debias observational estimates that have support in a larger region. These articles address covariate adjustment in observational data using standard causal inference methods that do not rely on instrumental variables. However, little methodological work has been done to combine IV analysis and adjustment for confounding effects. In the choice between IV analysis and standard causal inference, one must either utilize IVs and ignore any observed confounding factors or adjust for observed confounding factors and ignore any available IVs[1]. Often in real-world settings, some variables are IVs and some are confounders, and one could be interested in utilizing the entire data set to provide the best estimate for the CATE.

In this paper, inspired by the methodology developed to combine RCT and observational data, we will develop a novel method to combine the IV estimator with regression estimators adjusted for observed confounders to provide a consistent estimator of the CATE with relatively high efficiency. We will achieve this by treating IV analysis as target trials[8,9] and placing parametric restrictions on the residual unobserved confounding effect in the regression analysis. Furthermore, we will take advantage of the framework developed by Kazemi and Olsen[10] to generalize this method to allow for a categorical treatment variable. We illustrate the consistency and the relative efficiency of the estimator in a simulation study and apply the developed methodology to answer the clinical question presented in Subsection 1.2.

1.1 Background

Mogstad and Torgovitsky[11] provide a comprehensive review synthesizing modern IV literature that accounts for unobserved heterogeneity in treatment effects. They discuss marginal treatment effect analysis as a potential solution, treating the tendency to choose a treatment as a random continuous variable that can be affected by effect modifiers. With a continuous IV, the treatment effect can then be estimated as a function of this variable. However, this requires strong parametric assumptions or high instrumental variation.

Kallus et al.[12] develop a method to utilize experimental data to debias the observational estimate of the CATE outside the support of the experimental data. They propose parametric extrapolation of the unobserved confounding effect. Dimitriou et al.[13] expand on this approach by replacing parametric extrapolation with Gaussian processes. These methods do not account for non-compliance in the experimental data or that compliance can vary across the covariate space.

Güdemann et al.[14] propose triangulating confounder adjustment, instrumental variables, and difference-in-differences analysis, emphasizing that each method relies on distinct identifying assumptions and leverages different structure in the data. Their simulation study demonstrates that violations of assumptions can lead to substantial bias, making reliance on a single estimator fragile. They introduce a dissimilarity statistic that accounts for correlation between estimates and can formally assess whether different methods target the same underlying causal effect.

Oprescu and Kallus[15] propose a method to combine covariate adjustment and IV analysis for a dichotomous treatment using machine learning techniques for compliance-weighted bias correction, allowing varying but nonzero compliance across covariates. However, in practice, some subgroups may exhibit minimal or no compliance, making it difficult to reliably estimate treatment effects for these groups. Additionally, the application of machine learning for extrapolation in causal inference comes with interpretability and data volume limitations that restrict the transferability of the method to the medical field.

In an attempt to develop an IV framework for categorical treatment variables, Kazemi and Olsen[10] introduced the new concept of adherence set, defined as the set of relevant treatment alternatives for a decision team (treatment alternatives with positive probabilities of being chosen). Assuming monotonicity, there is a one-to-one correspondence between adherence sets and principal strata. In this way, the target population of the IV estimand is no longer defined based on counterfactual choices, but based on conditional treatment probabilities. However, the IV estimate is still valid only for the subpopulation where certain positivity conditions hold. We expand on the methodology developed by Kazemi and Olsen[10] and propose to utilize interpretable extrapolation techniques to estimate the CATE in low-compliance regions by combining instrumental and confounding information, still allowing categorical treatment variables.

1.2 Clinical Case

Tumor necrosis factor alpha inhibitors (TNFis) are a class of biologic drugs commonly used to treat rheumatoid arthritis (RA). There exist five structurally different TNFis, namely adalimumab (ADM), certolizumab pegol (CZP), etanercept (ETN), golimumab (GOM) and infliximab (INX). TNFis have previously been assumed to be equally effective in the treatment of RA. This assumption has recently been challenged through an observational study utilizing price as an IV in Norway[9,10], with LATEs for some head-to-head comparisons estimated to be significantly different from null. Although the target populations for head-to-head comparisons were thoroughly and transparently

described and discussed, previous work has stopped short of generalizing the results to the whole patient population due to heterogeneity concerns. Age is likely both a determinant of TNFi treatment choice and a plausible effect modifier[9]. Clinicians often consider age when selecting biologics because older patients have higher infection risks, more comorbidities, and may tolerate TNF inhibitors differently. At the same time, age-related differences in immune function and RA phenotype suggest that TNFi efficacy may vary across age groups[16,17]. In this paper, we estimate the CATEs as a function of age to investigate this potential heterogeneity.

2 Methods

For a categorical treatment, an ordinal IV and a continuous outcome, we will expand on the notation introduced by Kazemi and Olsen [10].

2.1 Notation & Assumptions

Let T, Y and Z be the treatment, outcome, and IV, respectively. Note that each value of the IV contains a complete ranking of all treatment alternatives based on level of encouragement. Let V, U and X be observed confounding variables, unobserved confounding variables, and observed effect modifying variables, respectively. This is formalized as the following structural equations and independence conditions:

$$T = f_T(Z, V, U, X, \delta), \tag{1}$$
$$Y = f_Y(T, V, U, X, \epsilon), \tag{2}$$
$$Z \perp\!\!\!\perp X, U, V \text{ and } \delta, \epsilon \perp\!\!\!\perp Z, V, U, X, \tag{3}$$

where δ and ϵ are residual variables.

A variable, X, is effect modifying if and only if:

$$\exists\, t, t' \in \text{Supp}(T) \mid Y^{T=t} - Y^{T=t'} \not\perp\!\!\!\perp X,$$

Note that given monotonicity, which we will assume, if $T \perp\!\!\!\perp X \Rightarrow$ LATE $=$ ATE. Therefore, moving forward, we focus on effect modifying variables that can also affect the treatment. Assume that all effect modifying variables that affect the choice of treatment are observed. This assumption is formalized as conditional homogeneity, defined as:

$$\forall\, t, t' \in \text{Supp}(T),\ Y^{T=t} - Y^{T=t'} \perp\!\!\!\perp Z, V, U, \delta \mid X. \tag{4}$$

This causal structure is presented as a directed graph in Figure 1.

We define the conditional average treatment response (CATR) as:

$$\theta_t(X) = E(Y^{T=t}|X).$$

The target parameter of interest is the CATE, defined as:

$$\tau_{t-t'}(X) = E(Y^{T=t} - Y^{T=t'}|X) = \theta_t(X) - \theta_{t'}(X). \tag{5}$$

Let $\omega_t(X)$ be a confounded estimate of the CATR, adjusted for V, but unadjusted for U. Let $\beta_t(X)$ be the bias:

$$\omega_t(X) = \int_v E(Y|V = v, X, T = t)dp(v|X)$$
$$\beta_t(X) = \omega_t(X) - \theta_t(X),$$

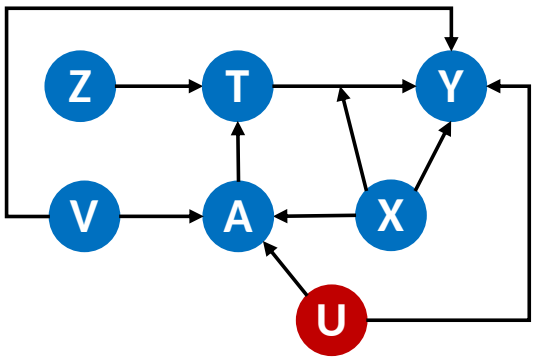


Figure 1. The directed acyclic graph defined by our assumptions. All blue variables are observed, while U represents unobserved confounding effects.

where $p(v|X)$ is the probability density of V given X. Assume that $\beta_t(X)$ has a known parametric form. In this paper, we assume that β_t is linear in X. However, this assumption can be easily relaxed.

Similarly to Kazemi and Olsen [10], we define the adherence set as:

$$A = \{t|P(T = t|V, U, X, \delta) > 0\}.$$

We assume the following choice function:

$$T = g(Z, A) = t_{\min(\{i|t_i \in A\})}, \text{ where } Z = (t_1, t_2, ..., t_{N_T})$$

assuming that decision teams always choose the most encouraged treatment (by the IV) that is in their adherence set. This assumption is referred to as monotonicity in the dichotomous case. Let N_Z and N_T be the number of observed values for the IV and available treatment alternatives, respectively. For any treatment, $t \in \text{Supp}(T)$, define:

$$P_Z(t, X) : N_Z \times 1, P_Z(t, X)[i] = P(T = t|Z = z_i, X)$$
$$Q_Z(t, X) : N_Z \times 1, Q_Z(t, X)[i] =$$
$$E(Y|Z = z_i, T = t, X)P(T = t|Z = z_i, X)$$
$$B_t : N_Z \times (2^{N_T} - 1),$$
$$B_t[i, n] = 1[T = t|Z = z_i, A = a_n],$$

where the indices i and n indicate the value of the IV and the adherence set, respectively. Let B_t^+ be the Moore–Penrose pseudo-inverse of B_t.

2.2 Identification & Estimation

Kazemi and Olsen [10] showed that for any binary solution b to the equation

$$b = bB_t^+ B_t, \tag{6}$$

the conditional local average treatment response (CLATR) is identified as:

$$\phi_t(b, X) = E(Y^{T=t}|b[A] = 1, X) = \frac{bB_t^+ Q_Z(t, X)}{bB_t^+ P_Z(t, X)},$$

where $b[A]$ is the value corresponding to the adherence set of a decision team in a solution to Equation 6. The condition $b[A] = 1$ characterizes decision teams with a combination of different adherence sets for which the CLATR is possible to identify given the observed instrumental values.

To increase the understandability of the proposed method, we will describe the different variables in the dichotomous case with the conventional language of IV analysis, namely principal strata (S). Assuming monotonicity, there is a one-to-one correspondence between principal strata and adherence set (A), the values of which can be always takers (AT: $A = \{1\} \Rightarrow T^{Z=0} = T^{Z=1} = 1$), never takers (NT: $A = \{0\} \Rightarrow T^{Z=0} = T^{Z=1} = 0$) and compliers (CL: $A = \{0, 1\} \Rightarrow T^{Z=0} = 0, T^{Z=1} = 1$). Equation 6 then has $M = 5$ solutions, identifying 6 CLATRs listed in Table 1.

The standard IV estimator estimates the effect for compliers ($S = CL$). If the distribution of effect modifiers (X) is substantially different in this subpopulation from the target population of the study, which should be defined with respect to the intervention in mind, the estimates will be biased with respect to the target parameter. This is equivalent to conducting an RCT in which the distribution of X is substantially different in the study population from the target population. The results of the RCT are therefore not generalizable to the target population. Furthermore, utilizing the standard conditional IV estimator to estimate the CATE is equivalent to using the data from this hypothetical RCT to estimate the conditional effect in regions of X where the RCT has little or no support, leading to high uncertainty. The joint probability distribution of X and compliance ($S = CL$) can be viewed as the probability of participation in the trial. This is given as:

$$\begin{aligned} f_{X,S=CL}(x) &= f_X(x)P(S = CL|X = x) \\ &= f_X(x)P(T = 1|Z = 1, X = x) \\ &\quad - f_X(x)P(T = 1|Z = 0, X = x). \end{aligned}$$

The density function of X for compliers is proportional to the joint distribution, namely:

$$f_{X|S=CL}(x) \propto f_{X,S=CL}(x). \tag{7}$$

Similarly, the conditional density function can be derived for always-takers and never-takers. In essence, we interpret the principal strata as two single-arm studies and one randomized two-arm study in the rest of this section and propose to model the bias of a confounded model such that IV estimates of CATRs are given more weight in regions of X where the density of X in the corresponding study is higher.

Lemma 1. *Under assumptions presented in Subsection 2.1, CATR and CLATR are equal up to a constant that depends only on b and X. Namely:*

$$\theta_t(X) = \phi_t(b, X) + C(b, X) \; \forall \, t, b \mid b = bB_t^+ B_t$$

Proof. Appendix Subsection A.1

Therefore, $\tau_{t-t'}(X)$ can, strictly speaking, be estimated by estimating ϕ_t and $\phi_{t'}$ for some common b and subtracting $\hat{\phi}_{t'}$ from $\hat{\phi}_t$. The challenge is that since A depends on X, there could be regions of X where the denominator of ϕ_t is diminishing for some treatments, leading to high variance.

Lemma 2. *Let the categorical IV (CIV) estimator be:*

$$\hat{\phi}_t(b, X) = \frac{\hat{U}}{\hat{V}}, \text{ where}$$
$$U = bB_t^+ Q_Z(t, X) \text{ and } V = bB_t^+ P_Z(t, X).$$

Then for large sample sizes, the variance of the CIV estimator can be approximated as:

$$\mathrm{Var}(\hat{\phi}_t(b, X)) \approx V^{-2}\mathrm{Var}(\hat{U} - \phi\hat{V}).$$

Proof. Appendix Subsection A.2

Let $b_1, b_2, ..., b_M$ be the set of all binary solutions to Equation 6 for all $t \in \mathrm{Supp}(T)$. We propose to model the difference between the CIV estimator and the confounded estimator with a weighted linear model with interaction terms. These solutions can be thought of as representing different single -or multiarm studies with high internal validity, and external validity that depends on X. The denominator of the CIV estimator can be interpreted as the probability of participation in the study given X. Assuming that $C(b_i, X)$s are linear in X, we regress $\hat{\omega}_t(X) - \hat{\phi}_t(b, X)$ linearly on X, T and b, including interaction terms TX and bX. We utilize inverse variance weighting to achieve higher efficiency[18]. Inspired by the variance expression given in Lemma 2, the target trial argument in Equation 7 and the fact that the variance of any local estimator is inversely proportional to the density of X[19], we weight the observations with the square of the denominator of the CIV estimator times the density of X. Namely:

$$\begin{aligned} \hat{\omega}_t(X) - \hat{\phi}_t(b, X) = \alpha_t + \alpha_{t,X}X + \\ \sum_{i=1}^{M}(\alpha_i + \alpha_{b_i}X)1[b = b_i] + \gamma, \text{ where} \\ \gamma \sim \mathcal{N}\left(0, \left(\frac{\sigma}{bB_t^+\hat{P}_Z(t, X)}\right)^2 \cdot \frac{1}{\hat{f}_X(X)}\right). \end{aligned} \tag{8}$$

Note that this weighting neither takes into account the variance of $\hat{\omega}_t$ nor includes all terms in the variance of $\hat{\phi}_t$. The variance of $\hat{\omega}_t$ obviously depends on the specific data structure and the estimating model, but in practice it is likely negligible compared to the variance of $\hat{\phi}_t$. Given unbiased estimation for P_Z and Q_Z, a full approximate expression for the variance of $\hat{\phi}_t$ can be derived through Lemma 2 and utilized in the weighting to achieve even more efficient estimation. However in practice, for IVs with X-dependent strength, the variance is dominated by the V^{-2} term in most regions of X. We argue, therefore, that the weights provided in model 8, while simple and easy to calculate, are a good approximation to the optimal weights.

For any pair of treatments, (t, t'), we can construct a debiased CATE estimator.

Theorem 1. *If both $\beta_t(X)$ and $C(b, X)|b$ are linear in X, then $\tau_{t-t'}(X)$ can be consistently estimated from the model presented in Equation 8 as:*

$$\hat{\tau}_{t-t'}(X) = \hat{\omega}_t(X) - \hat{\omega}_{t'}(X) - \hat{\alpha}_t + \hat{\alpha}_{t'} - (\hat{\alpha}_{t,X} - \hat{\alpha}_{t',X})X$$

We refer to this estimator as the heterogeneous IV (HIV) estimator.

Proof. Appendix Subsection A.3

Next to generalization to allow for categorical treatment, we make three main contributions to methods for estimating the CATE:

i	CLATR	Response estimator	Probability estimator
1	$E(Y^{T=0}\|S=NT,X)$	$\hat{E}_1=\hat{E}(Y\|T=0,Z=1,X)$	$\hat{P}_1=\hat{P}(T=0\|Z=1,X)$
2	$E(Y^{T=0}\|S\in\{NT,CL\},X)$	$\hat{E}_2=\hat{E}(Y\|T=0,Z=0,X)$	$\hat{P}_2=\hat{P}(T=0\|Z=0,X)$
3	$E(Y^{T=0}\|S=CL,X)$	$\frac{\hat{E}_2\hat{P}_2-\hat{E}_1\hat{P}_1}{\hat{P}_2-\hat{P}_1}$	$\hat{P}_2-\hat{P}_1$
4	$E(Y^{T=1}\|S=AT,X)$	$\hat{E}_4=\hat{E}(Y\|T=1,Z=0,X)$	$\hat{P}_4=\hat{P}(T=1\|Z=0,X)$
5	$E(Y^{T=1}\|S\in\{AT,CL\},X)$	$\hat{E}_5=\hat{E}(Y\|T=1,Z=1,X)$	$\hat{P}_5=\hat{P}(T=1\|Z=1,X)$
3	$E(Y^{T=1}\|S=CL,X)$	$\frac{\hat{E}_5\hat{P}_5-\hat{E}_4\hat{P}_4}{\hat{P}_5-\hat{P}_4}$	$\hat{P}_5-\hat{P}_4$

Table 1. The first column, i, refers to the indices of b_is, which are solutions of Equation 6, each corresponding to a target population defined by the condition on S. The second column includes all identifiable conditional local average treatment responses (CLATRs). The third column shows how the corresponding CLATR can be estimated with observables. The last column shows an estimator of the conditional probability of being in the target population. $\hat{E}_i$s and $\hat{P}_i$s are used as shorthand notation to present the estimators for compliers in a simpler way.

- Combining IV and covariate adjustment for more accurate inference of nonlinear CATEs.
- Variance weighting for more precise estimation.
- Utilizing all IV identifiables instead of only the complier effect to further increase precision.

3 Simulation Study

To illustrate the power of the method presented in Section 2 to estimate the CATE, we present a simple simulation study with dichotomous treatment and confounding variables, continuous effect modifying and outcome variables, and quadratic effect modification. The setup is as follows:

$$Z,U,V\sim\mathcal{B}(n,0.5);\epsilon,\delta\sim\mathcal{N}(0,1)\text{ and }X\sim\mathcal{N}(2,1)$$
$$p=\text{logit}^{-1}(3Z-3X+2U+V+\delta)\text{ and }T\sim\text{Bernoulli}(p)$$
$$Y=(X^2+X+1)T-8U-V+X+XU+\epsilon$$

Given this setup, the true CATE is given as:

$$\tau_{1-0}(X)=X^2+X+1.$$

Note that in this setup, the confounding, β, is not linear in X due to the logistic link. Still, we will show that modeling it with a linear regression model outperforms other competing methods.

We estimate $P_Z(t,X)$ with a logistic regression with T as the outcome, regressed on Z and X without any interaction effects. We estimate $Q_Z(t,X)$ by combining the model for $P_Z(t,X)$ with a linear regression model with Y as the outcome, regressed on Z,X and T, including the interaction effects ZX and TX. Note that these are not local estimators. The variance expression in Lemma 2 is, therefore, not necessarily theoretically valid. Practically however, we apply the weights obtained from this lemma. For the confounded estimator, $\omega_t(X)$, we fit a generalized additive model (GAM) to Y with group-specific smooth splines in X by T and linear terms for the main effects of T,V,X and the interaction term VX. Lemma 2 gives the CIV estimate together with the weights to be used to estimate the bias. Once we have $\hat{\omega}$ and $\hat{\phi}$, we will use Theorem 1 and the model presented in Equation 8 to estimate τ.

Repeating this experiment 100 times for the number of observations, n, ranging from 1000 to 5000 with steps of 1000, we provide an estimate of the root mean squared error (RMSE) as follows:

$$\text{RMSE}(n)=\left(\frac{1}{100}\sum_{\text{sim}=1}^{100}\int_X(\hat{\tau}_{1-0}(X)-\tau_{1-0}(X))^2dp(X)\right)^{\frac{1}{2}},$$

where $p(X)$ is the density function of X.

Inspired by earlier literature[15,20], we consider the following four competing methods:

- Standard conditional IV estimator given as:

  $$\hat{\tau}_{IV}(X)=\frac{\hat{E}(Y|Z=1,X)-\hat{E}(Y|Z=0,X)}{\hat{P}(T=1|Z=1,X)-\hat{P}(T=1|Z=0,X)},$$

  We estimate the numerator by regressing Y linearly on X,Z and ZX, while the denominator is estimated with a logistic regression model as for the HIV estimator.
- The confounded estimator,

  $$\hat{\tau}_S(X)=\hat{\omega}_1(X)-\hat{\omega}_0(X),$$

  which adjusts for observed confounding effects by fitting a GAM to Y as for the HIV estimator.
- A weighted IV estimator, $\hat{\tau}_W(X)$, obtained by fitting a linear regression model to $\hat{\tau}_{IV}(X)$ including X and X^2 and weighted with the density of X times the square of the denominator in $\hat{\tau}_{IV}$. Note that the parametric form of the CATE is normally unknown. Still, we will show the inferiority of this approach even if the parametric form is known.
- A combined estimator, $\hat{\tau}_C(X)$, obtained by fitting a linear regression model to the bias $\hat{\tau}_S(X)-\hat{\tau}_{IV}(X)$, weighted with the density of X times the square of the denominator in $\hat{\tau}_{IV}$. Note that the only difference between this and the HIV estimator, $\hat{\tau}_{1-0}(X)$, is that $\hat{\tau}_{IV}(X)$ and therefore $\hat{\tau}_C$ only rely on the CLATE for compliers, while the HIV estimator utilizes the CLATRs for never and always takers as well.

Note that none of the competing IV methods presented here can easily be generalized to the case of categorical treatment. We show that even for dichotomous treatment, the competing IV methods do not perform well in estimating the CATE in the presence of effect heterogeneity. The result of the

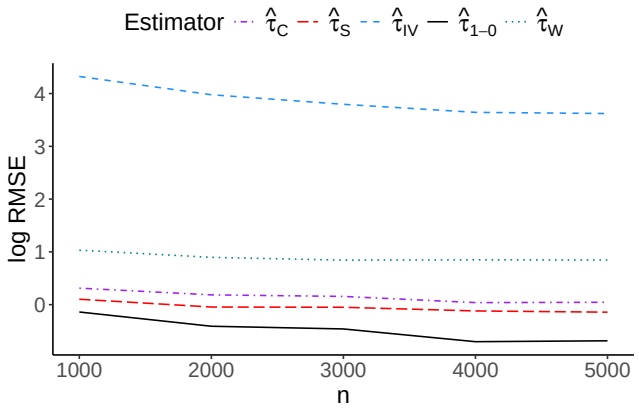


Figure 2. The log RMSE of the HIV estimator ($\hat{\tau}_{1-0}$) together with the log RMSEs for all competing methods. Since the RMSE of the standard IV method was substantially higher than that of the other methods, the RMSE has been log-transformed for readability purposes. The x-axis, n, is the number of observations in the simulated data.

simulation study is presented in Figure 2. One can see that even in the presence of a large unobserved confounding effect, all competing IV estimators are outperformed by the conventional estimator, namely applying GAM to only adjust for the observed confounding effect. The only IV method that outperforms covariate adjustment is the HIV estimator.

The conditional and marginal density functions of X in this simulation study are presented in Figure 3. In this figure, one can see that the IV estimator performs better than covariate adjustment in the high-compliance region of X. However, it is substantially outperformed outside this region. Its poor overall performance is exacerbated because the marginal density of X and the conditional density of X for compliers are different.

This simulation study serves an illustrative rather than a confirmatory purpose, and the comparative performance of the methods naturally depends on the simulation parameters. The performance of the IV estimators deteriorate as effect modification increases, while the performance of the covariate adjusted estimator deteriorates as unobserved confounding strengthens. Traditionally, the choice between these methods has been guided by which untestable assumption, homogeneity or no unobserved confounding, seems more plausible. The HIV estimator alleviates the need to rely on either assumption by combining the strengths of both approaches.

4 Clinical Application

We will demonstrate the applicability of the method developed in this paper on the data set analyzed by Kazemi et al.[9]. This data set contains clinical data on 1446 RA patients initiating a TNFi as their first biologic treatment in the period $2010-2023$. In this period, price was the main determinant of the first-line biologic treatment for RA patients, and can therefore be utilized as an IV. The IV assumptions have previously been sufficiently justified[9,10].

Kazemi et al.[9] applied an IV analysis and a regression analysis that adjusted for baseline variables to compare TNFis, using clinical remission at 3 months as the outcome. The results of the two analyses do not agree. Potential reasons for the disagreement are unobserved confounding effects and effect heterogeneity. Age was suggested as a potential effect modifier. In this paper, we combine regression and IV analysis to estimate the CATE as a function of age.

We will use the disease activity score (DAS28 CRP) at 3 months as the outcome. This is a continuous composite score ranging from $0-9.4$, including patient reported and clinical outcomes. A lower score means better response to the treatment. Missing data is handled with multiple (10) imputations with chained equations[21] and uncertainty is estimated with bootstrapping. Similarly to the primary analysis performed by Kazemi et al.[9], we limit possible adherence sets to 14 clinically meaningful sets. We use a GAM as the confounded estimator with smooth splines in age by treatment and include interaction terms between age and all observed baseline variables with missingness less than 70%. To estimate the denominator of the CIV estimator, we use a multinomial regression model of treatment regressed on sorted price list and age. For the numerator, we include all interaction effects between the price list, the treatment, and age. The density function for age is estimated with kernel density estimation[22].

The results are presented in Figure 4. One can see that the LATEs estimated by Kazemi et al.[9] remain mostly valid across age groups, suggesting that age does not substantially modify the effects. However, the curves tend to converge for older patients, indicating smaller differences between treatments in this group. The inferiority of GOM is more apparent among younger patients. In contrast, all confidence intervals exclude null difference compared to ADM in older patients, suggesting its inferiority in this subgroup.

5 Discussion & Conclusion

In this paper, we propose a novel framework for estimating CATEs for a categorical treatment when both IVs and observed confounding variables are present. Our approach builds on recent developments in combining information from randomized and observational data, extending this line of work to the IV setting. Specifically, we show that by interpreting the IV analysis in a target trial emulation framework and modeling the residual bias in the confounded estimator, one can exploit both the IV and observed covariates to obtain a consistent and more precise CATE estimator. This approach allows generalization of IV results to a larger population when treatment effects are heterogeneous.

A common critique of observational studies is that they are inherently confounded, as if confounding were an immutable property of nonrandomized data. However, as Hernán and Robins[23] emphasize, confounding is a scientific rather than a philosophical problem: if one posits that an association is confounded, one must specify the confounder and its causal pathway so that it can, in principle, be measured and adjusted for. The same standard should apply to criticisms of IV approaches. Dismissing IV methods on the grounds that LATE lacks interpretability is no more scientific than rejecting all observational studies as confounded. If one believes that the complier population differs significantly in

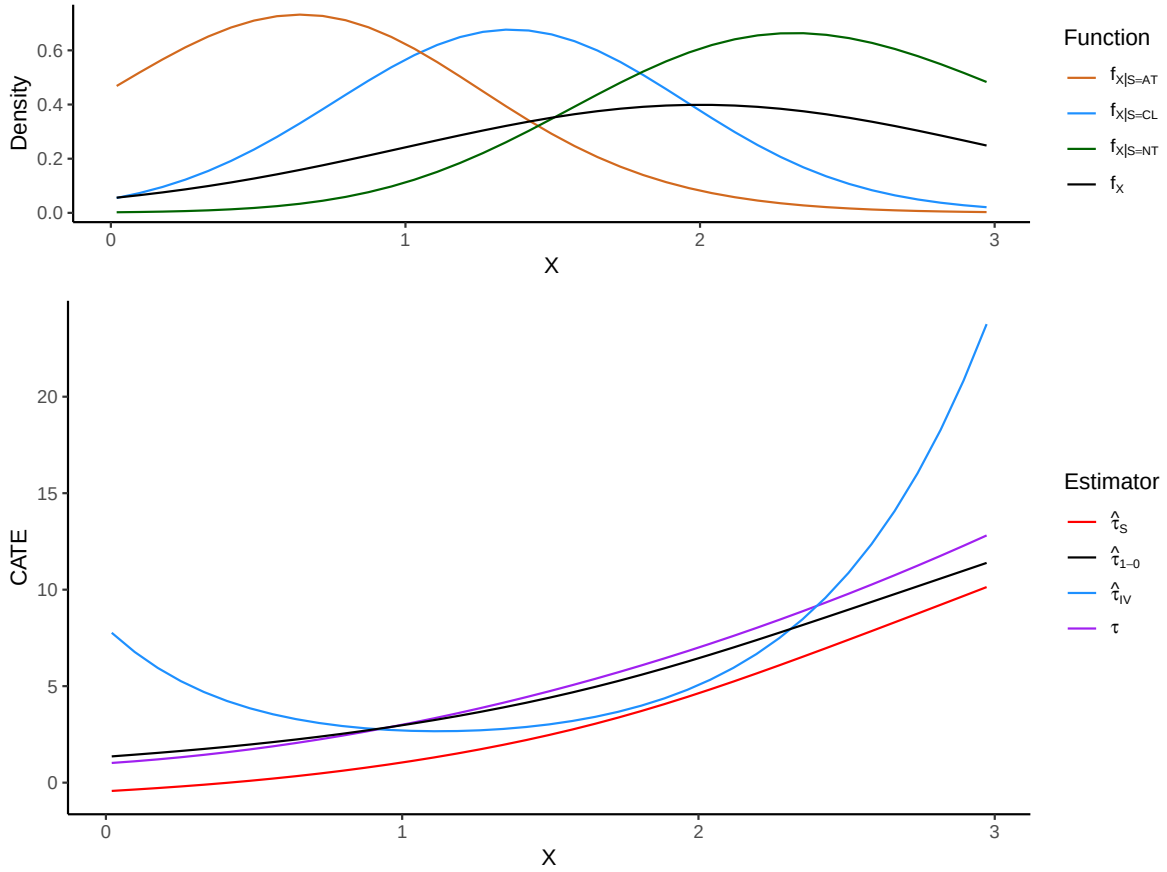


Figure 3. In the top plot, the marginal and conditional density functions of X for each principal stratum in this simulation study are presented. In the bottom plot the results of one simulation are shown with $n = 5000$ observations. For simplicity, $\hat{\tau}_W$ and $\hat{\tau}_C$ are excluded from this plot as it serves an illustrative purpose.

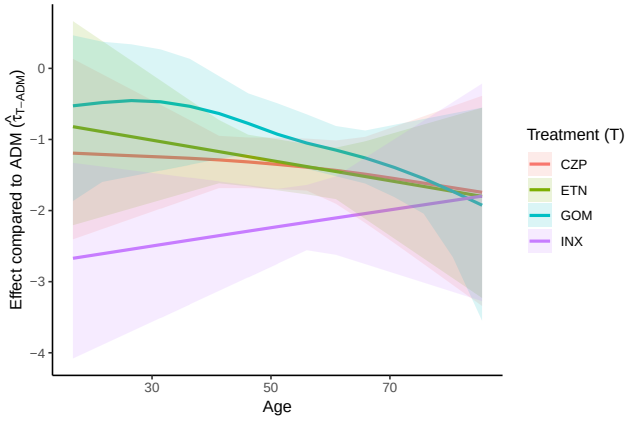


Figure 4. Estimated average effect of each TNFi compared to the most common one (adalimumab) conditional on age. Lower effect means better response to the treatment. Confidence intervals are calculated using bootstrapping with 100 replications. Given conditional homogeneity, the effect curves can also be compared to each other.

treatment effect from the target population, then one should specify the effect modifiers that induce this heterogeneity. Otherwise, the assumption of conditional homogeneity remains plausible. Once effect modifying variables are specified and observed, an estimator of the CATE is needed given conditional homogeneity. In this paper, by combining instrumental and covariate information, we have provided the methodological tool for estimating the CATE for a categorical treatment in the presence of observed effect modifying factors that also affect the choice of treatment.

Through our simulation study, we demonstrated that the HIV estimator can outperform competing IV-based methods in terms of RMSE. We illustrate that even though our proposed weights for the bias model are not optimal, the resulting estimate matches the true CATE curve to a high degree in a realistic setting. Even higher performance can be achieved by using better approximations for the variance in the weights. Importantly, the simulation also shows that covariate adjustment methods, although confounded, can sometimes outperform conditional IV estimators in finite samples when effect heterogeneity is substantial and compliance is limited. The HIV estimator combines the strengths of both approaches.

The clinical application to the comparative effectiveness of TNFis in RA illustrates the applicability of our method. Detailed clinical discussion of the results is outside the scope of this work. However, it should be noted that while previous work identified significant LATEs using price as an IV, the interpretability of these estimates was limited to specific subpopulations defined by adherence. In contrast, our approach estimates treatment effects as a function of age, increasing clinical interpretability. The results suggest modest heterogeneity in treatment effects by age, with smaller between-drug differences among older patients. This provides an important refinement of previous findings

and offers a more generalizable foundation for treatment recommendations in clinical practice. Furthermore, ADM has been absent from earlier comparisons due to the denominator of the CIV estimator being too small, meaning that any LATE identified involving this drug would only be valid for a small subpopulation of patients. However, with our approach, ADM is reliably compared to the other drugs.

Our approach relies on certain structural and modeling assumptions. First, we assume that all effect modifiers that can also affect the choice of treatment are observed and that the residual bias function is linear. Although the simulation suggests robustness to moderate violations of linearity, the performance of the HIV estimator compared to competing methods under strong nonlinearity or when some effect modifiers are unobserved remains an open question. Second, as in all IV analyses, the validity of the IV is crucial. Any direct effect of the IV on the outcome or strong violations of the assumed choice function (monotonicity) may lead to bias. Future work could explore the use of flexible, nonparametric approaches to model the bias function or to incorporate multiple IVs, potentially improving robustness in complex real-world settings.

Data and Code Availability

The analyses were performed in R version 4.2.1. All code, anonymous data from the NOR-DMARD registry, all generated data sets and further information are available in the Open Science Framework (OSF) repository at:

https://osf.io/dtcgh

Contributions Statement & Competing Interests

No competing interest is declared.

Acknowledgements

The authors thank Dr. Eirik Klami Kristiandslund and Dr. Sella Provan for valuable clinical discussions, in addition to the participants, study leaders and personnel of NOR-DMARD.

Funding

This work is supported by funds from the South-Eastern Norway Regional Health Authority (HSØ: # 2021024).

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Supplemental material

A Proofs

A.1 Proof of Lemma 1

If $b = bB_t^+ B_t$ only for a single $t = t_0$, it is obvious that:

$$\theta_{t_0}(X) - \phi_{t_0}(b, X) = C(b, X).$$

If $b = bB_t^+ B_t$ for more than one t, pick any default treatment, $t_0 \in \text{Supp}(T)|b = bB_{t_0}^+ B_{t_0}$. Then by definition, for any other $t \in \text{Supp}(T)|b = bB_t^+ B_t$:

$$\theta_t(X) = \tau_{t-t_0}(X) + \theta_{t_0}(X) \tag{9}$$
$$\phi_t(b, X) = \phi_{t_0}(b, X) + E(Y^{T=t} - Y^{T=t_0}|b[A] = 1, X). \tag{10}$$

Since $Y^{T=t} - Y^{T=t_0} \perp\!\!\!\perp V, U, \delta|X$ by Assumption 4 and $b[A]$ is a function of A, which itself is a function of V, U, X and δ, it can be inferred that $Y^{T=t} - Y^{T=t_0} \perp\!\!\!\perp b[A]|X$. Therefore:

$$E(Y^{T=t} - Y^{T=t_0}|b[A] = 1, X) = \tau_{t-t_0}(X).$$

Equations 9 and 10 can then be rewritten as:

$$\theta_t(X) = \tau_{t-t_0}(X) + \theta_{t_0}(X) \tag{11}$$
$$\phi_t(b, X) = \tau_{t-t_0}(X) + \phi_{t_0}(b, X). \tag{12}$$

Subtract Equation 11 from one Equation 12:

$$\theta_t(X) - \phi_t(b, X) = \theta_{t_0}(X) - \phi_{t_0}(b, X) = C(b, X). \quad \square$$

A.2 Proof of Lemma 2

Let:

$$U := bB_t^+ Q_Z(t, X), \quad V := bB_t^+ P_Z(t, X), \quad \phi = \frac{U}{V}$$
$$\hat{U} := bB_t^+ \hat{Q}_Z(t, X), \quad \hat{V} := bB_t^+ \hat{P}_Z(t, X), \quad \hat{\phi} = \frac{\hat{U}}{\hat{V}}.$$

Applying the delta method for $f(U, V) = U/V$:

$$\text{Var}(\hat{\phi}) \approx \left(\frac{\partial f}{\partial U}\right)^2 \text{Var}(\hat{U}) + \left(\frac{\partial f}{\partial V}\right)^2 \text{Var}(\hat{V}) + 2\frac{\partial f}{\partial U}\frac{\partial f}{\partial V}\text{Cov}(\hat{U}, \hat{V}), \text{ with}$$
$$\frac{\partial f}{\partial U} = \frac{1}{V} \text{ and } \frac{\partial f}{\partial V} = -\frac{U}{V^2}.$$

Substituting gives

$$\begin{aligned}\text{Var}(\hat{\phi}) &\approx \frac{\text{Var}(\hat{U})}{V^2} + \frac{U^2\text{Var}(\hat{V})}{V^4} - 2\frac{U}{V^3}\text{Cov}(\hat{U}, \hat{V}) \\ &= \frac{1}{V^2}(\text{Var}(\hat{U}) + \frac{U^2\text{Var}(\hat{V})}{V^2} - 2\frac{U}{V}\text{Cov}(\hat{U}, \hat{V})) \\ &= \frac{1}{V^2}\text{Var}(\hat{U} - \phi\hat{V}).\end{aligned}$$

A.3 Proof of Theorem 1

First, we prove that the model presented in Equation 8 is correctly specified to estimate $\omega - \phi$. First, assume that $b = bB_t^+ B_t$, then:

$$\begin{aligned}&\omega_t(X) - \phi_t(b, X) = \\ &\quad \omega_t(X) - \theta_t(X) + \theta_t(X) - \phi_t(b, X) = \\ &\quad \beta_t(X) + \theta_t(X) - \phi_t(b, X) = \\ &\quad \beta_t(X) + C(b, X),\end{aligned}$$

where the second equation comes from the definition of β and the third equation comes from Lemma 1. Since both $\beta_t(X)$ and $C(b, X)$ are linear in X, $\omega_t(X) - \phi_t(b, X)$ is also linear in X. If $b \neq bB_t^+ B_t$, Lemma 1 cannot be directly applied to infer the third equation. However, we know there must exist another $t' \in \text{Supp}(T)$ such that $b = bB_{t'}^+ B_{t'}$, then:

$$\begin{aligned}&\omega_t(X) - \phi_t(b, X) = \\ &\quad \beta_t(X) + \theta_t(X) - \phi_t(b, X) = \\ &\quad \beta_t(X) + (\theta_{t'}(X) - \phi_{t'}(b, X)) - \\ &\quad (\theta_{t'}(X) - \phi_{t'}(b, X)) + \theta_t(X) - \phi_t(b, X) = \\ &\quad \beta_t(X) + C(b, X) + \tau_{t-t'} - (\phi_t(b, X) - \phi_{t'}(b, X)) = \\ &\quad \beta_t(X) + C(b, X) + \tau_{t-t'} - \tau_{t-t'} = \\ &\quad \beta_t(X) + C(b, X),\end{aligned}$$

where the third equation comes from Lemma 1 and the definition of τ and the fourth equation comes from Assumption 4. We have now shown that given b, $\omega_t(X) - \phi_t(b, X)$ is linear in X. The model in Equation 8 is, therefore, correctly specified. It remains to show that τ can be estimated from this model.

Define:

$$\begin{aligned}C'(X) &= \frac{\sum_{i=1}^M C(b_i, X)P(b = b_i|X)}{\sum_{i=1}^M P(b = b_i|X)} \\ \beta'_t(X) &= \beta_t(X) + C'(X) \\ \theta'_t(X) &= \theta_t(X) - C'(X),\end{aligned}$$

where the conditional probability of $b|X$ is given as:

$$P(b = b_i|X) = bB_t^+ P_Z(t, X) \; \forall \; t \mid b = bB_t^+ B_t.$$

. To prove the theorem, we will show:

$$\begin{aligned}&\hat{\tau}_{t-t'}(X) = \hat{\theta}'_t(X) - \hat{\theta}'_{t'}(X), \text{ where} \\ &\quad \hat{\theta}'_t(X) = \hat{\omega}_t(X) - \hat{\beta}'_t(X) \text{ and} \\ &\quad \hat{\beta}'_t(X) = \hat{\alpha}_t + \hat{\alpha}_{t,X}X + \frac{\sum_{i=1}^M(\hat{\alpha}_i + \hat{\alpha}_{b_i}X)\hat{P}(b = b_i|X)}{\sum_{i=1}^M \hat{P}(b = b_i|X)}.\end{aligned}$$

First, we prove the first two equations using only the definitions of θ' and β':

$$\begin{aligned}\tau_{t-t'}(X) &= \theta_t(X) - \theta_{t'}(X) \\ &= \theta_t(X) - C'(X) - (\theta_{t'}(X) - C'(X)) \\ &= \theta'_t(X) - \theta'_{t'}(X) \\ \theta'_t(X) &= \theta_t(X) - C'(X) \\ &= \omega_t(X) - \beta_t(X) - C'(X) \\ &= \omega_t(X) - \beta'_t(X).\end{aligned}$$

Now it remains to show:

$$\begin{aligned}\beta'_t(X) &= \frac{\sum_{i=1}^M(\omega_t(X) - \phi_t(b, X))P(b = b_i|X)}{\sum_{i=1}^M P(b = b_i|X)} \\ &= \alpha_t + \alpha_{t,X}X + \frac{\sum_{i=1}^M(\alpha_i + \alpha_{b_i}X)P(b = b_i|X)}{\sum_{i=1}^M P(b = b_i|X)},\end{aligned} \tag{13}$$

where the second equation can be inferred from the fact that the model is correctly specified. It remains to prove the first equation. First, we need to show that for any $b \in \{b_1, b_2, ..., b_M\}$:

$$\beta_t(X) = \omega_t(X) - \phi_t(b, X) - C(b, X). \tag{14}$$

If $b = bB_t^+ B_t$, this follows directly from the definition of β and Lemma 1. If $b \neq bB_t^+ B_t$, we know that there exists a t' such that $b = bB_{t'}^+ B_{t'}$. Then:

$$\begin{aligned}
\beta_t(X) &= \omega_t(X) - \theta_t(X) \\
&= \omega_t(X) - \theta_t(X) + \theta_{t'}(X) - \theta_{t'}(X) \\
&= \omega_t(X) - \tau_{t-t'}(X) - \theta_{t'}(X) \\
&= \omega_t(X) - \tau_{t-t'}(X) - (\phi_{t'}(b, X) + C(b, X)) \\
&= \omega_t(X) - \tau_{t-t'}(X) - (\phi_t(b, X) - \tau_{t-t'}(X) + C(b, X)) \\
&= \omega_t(X) - \phi_t(b, X) - C(b, X),
\end{aligned}$$

where the third equation follow from the definition of τ, the fourth equation from Lemma 1 and the fifth equation from Assumption 4. Taking a weighted average of Equation 14 over all values of $b \in \{b_1, b_2, ..., b_M\}$, will give equation 13, and the proof is complete. □